ROAD DESIGN - PRELIMINARY DESIGN DRAWINGS

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Standards profile: US - AASHTO Green Book 2018 / FHWA Design speed 50 km/h

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GENERAL NOTES

1. Design to US - AASHTO Green Book 2018 / FHWA - parameter register lists each value as verified (from the published standard), formula, or engineer input.
2. Values from copyrighted or paywalled tables are not shipped with the software; they are entered by the engineer and marked as engineer inputs in the parameter register and design report.
3. Dimensions in metres unless noted; elevations in metres on the survey datum; stations in km+m along the control line.
4. Existing ground from the supplied survey/terrain points (TIN). Verify on site before construction.
5. Earthwork volumes by average end area with a composite-grid cross check; shrink/bulk factors and haul limits as stated in the earthwork report.
6. Standard details are schematic references; construct to the cited agency standard drawings and specification.
7. These drawings are generated by agent tools from the audit-logged design. Status: PRELIMINARY unless the engineer has recorded approval; sealing is the engineer's act.
8. Selected route: A-direct (recorded with reasons in the audit log).

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cover, sheet index and general notes

Sheet 1 of 19

Scale: NTS (A1)

Standards: Us

Design speed: 50 km/h

Checks: 218/218 pass

Audit log head:

c340436ef0da561e02ee22ad660b07ad

a4df5d59149906780694e48e0e9c0ea5

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

REVIEWER-PLACEHOLDER

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Alternative	Length m	Min R m	Max grade %	Cut m3	Fill m3	Imbalance m3	Wall conc. m3	Mandatory fails	Cost proxy
A-direct	1136	180	8.00	22618	51310	-30954	6581	0	317776
B-contour-following	1088	85	8.00	39482	69081	-33548	4691	3	299615
C-separate-corridor	1129	185	8.00	20967	100404	-81533	77824	0	2578395

Selection rule: fewest mandatory geometry failures, then lowest cost proxy (unit rates are agent placeholders unless the engineer enters rates)
- vs B-contour-following: cost proxy 317776 vs 299615 (+6.1%), length 1136 vs 1088 m, cut+fill 73928 vs 108563 m³, imbalance -30954 vs -33548 m³, wall concrete 6581 vs 4691 m³, max grade 8.0 vs 8.0 %, mandatory failures 0 vs 3
- vs C-separate-corridor: cost proxy 317776 vs 2578395 (-87.7%), length 1136 vs 1129 m, cut+fill 73928 vs 121371 m³, imbalance -30954 vs -81533 m³, wall concrete 6581 vs 77824 m³, max grade 8.0 vs 8.0 %, mandatory failures 0 vs 0
Cost proxy uses placeholder unit rates unless the engineer enters rates; it ranks alternatives and is not an estimate. Least-cost search on a 10 m grid, 16 headings, grade penalty above the profile maximum.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Route alternatives and selection
Sheet 2 of 19
Scale: 1:2500 (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
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a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
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PRELIMINARY

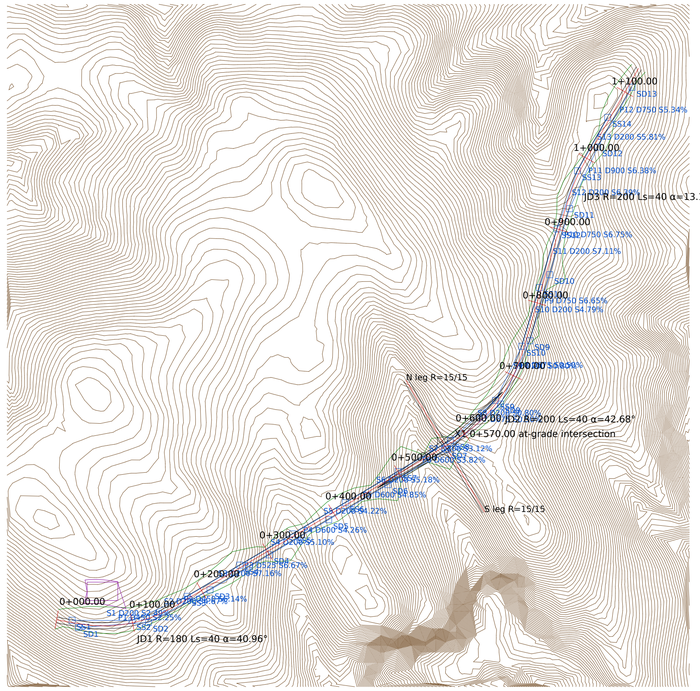
NOT FOR CONSTRUCTION

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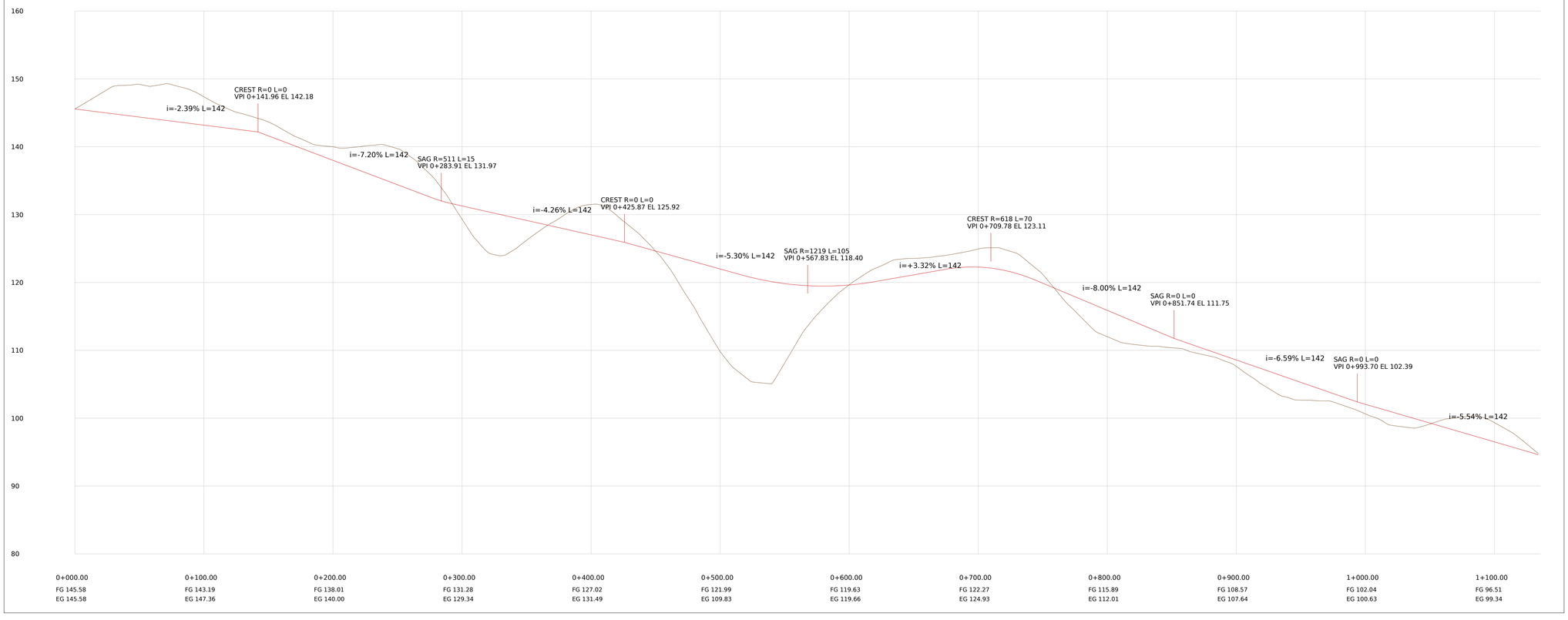
Not reviewed or sealed by a licensed engineer
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PLAN 1:4000 contours 1 m (index 5 m) N↑



PROFILE H 1:1829 V 1:348 (existing ground brown, finished grade red)



Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Plan and profile

Sheet 3 of 19

Scale: plan 1:4000; profile as shown (A1)

Standards: Us

Design speed: 50 km/h

Checks: 218/218 pass

Audit log head:
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Prepared by: civil-mcp agent tools

Engineer in responsible charge:
REVIEWER-PLACEHOLDER

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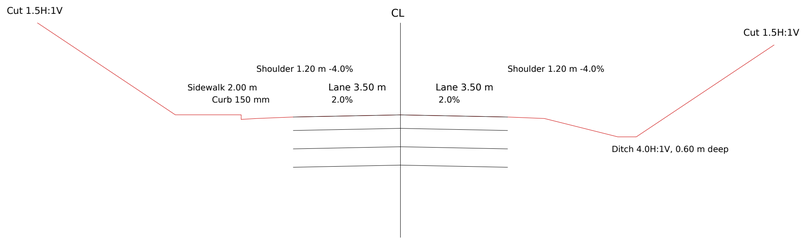
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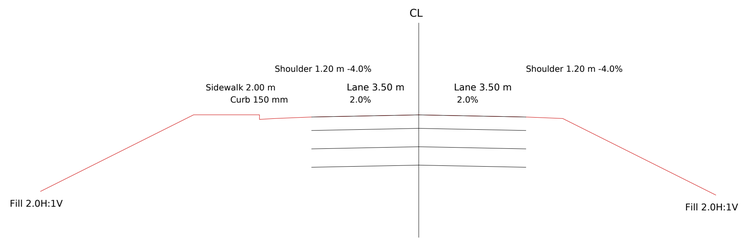
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TYPICAL SECTION IN CUT



TYPICAL SECTION IN FILL



Layer (top to bottom)	Thickness mm
Asphalt concrete	110
Aggregate base	150
Aggregate subbase	150

Pavement layer lines are drawn at 4x vertical exaggeration for legibility. Side slopes shown for a nominal 3.0 m cut / 2.5 m fill height; actual daylight per cross sections. Barrier, lining and protection treatments per the protection layout sheet.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Typical sections

Sheet 4 of 19

Scale: 1:100 (A1)

Standards: Us

Design speed: 50 km/h

Checks: 218/218 pass

Audit log head:

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Prepared by: civil-mcp agent tools

Engineer in responsible charge:

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PRELIMINARY

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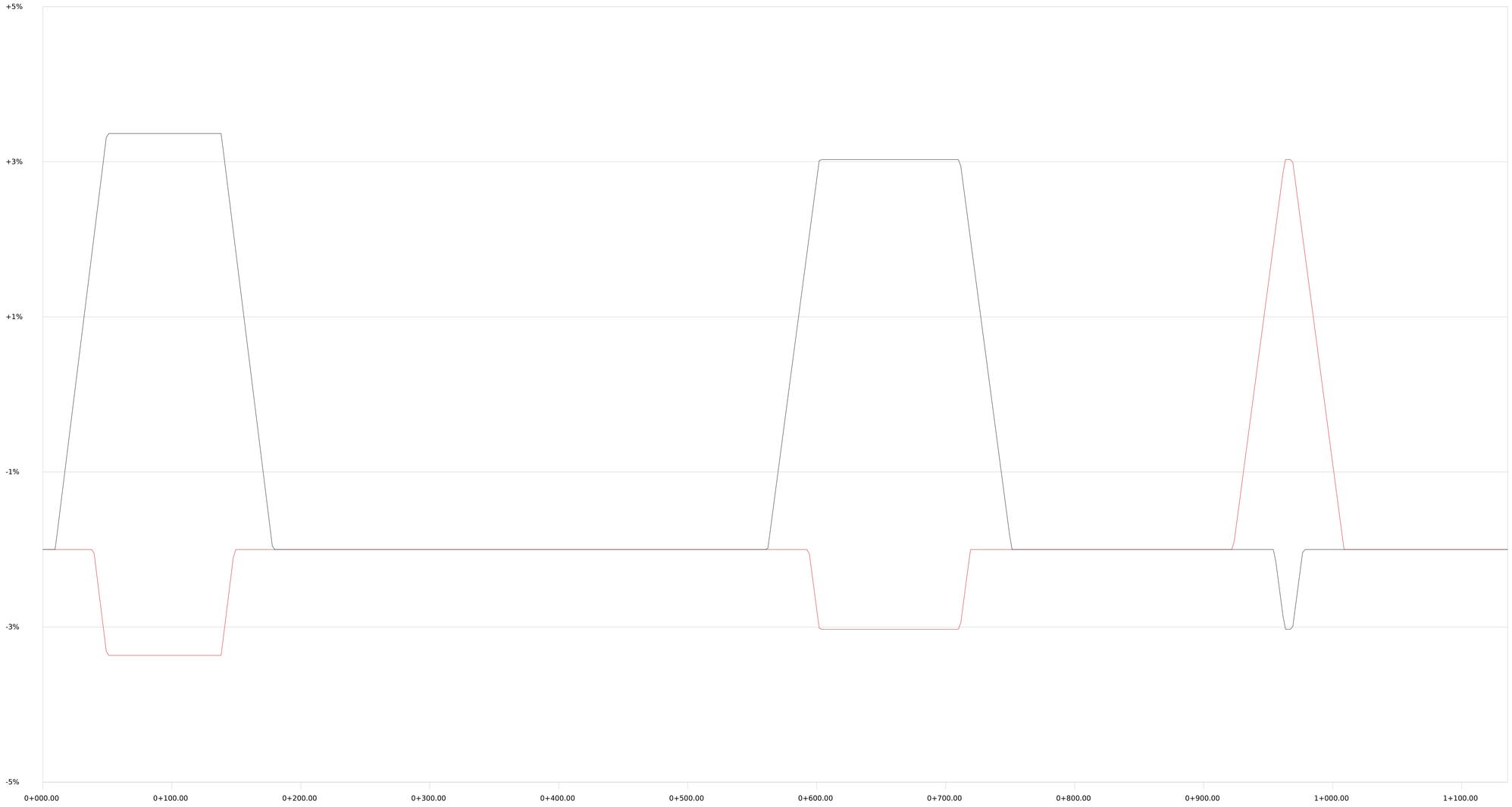
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Right edge cross slope
Left edge cross slope (rise per run outward from CL)



Curve	R m	e	Method	Runoff start	Full e	Full e end	Runoff end	Rel. gradient %
1	180	0.034	proportional distribution (AASHTO Method 1): e = emax·Rmin/R = 0.034·175.0...	0+049.58	0+049.58	0+138.25	0+178.25	0.469
2	200	0.030	proportional distribution (AASHTO Method 1): e = emax·Rmin/R = 0.030·175.0...	0+562.02	0+602.02	0+711.01	0+751.01	0.440
3	200	0.030	proportional distribution (AASHTO Method 1): e = emax·Rmin/R = 0.030·175.0...	0+922.83	0+962.83	0+968.80	1+008.80	0.440

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Superelevation diagram
Sheet 5 of 19
Scale: H 1:as shown; V as shown (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
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Prepared by: civil-mcp agent tools
Engineer in responsible charge:
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PRELIMINARY

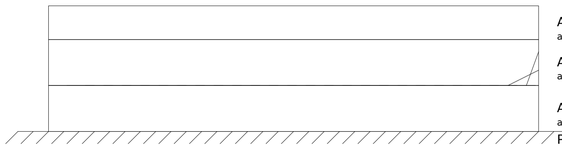
NOT FOR CONSTRUCTION

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PAVEMENT STRUCTURE (FLEXIBLE)



Asphalt concrete - 110 mm
a = 0.44, m = 1

Aggregate base - 150 mm
a = 0.14, m = 1

Aggregate subbase - 150 mm
a = 0.11, m = 1

Prepared subgrade, design CBR 6%

Total thickness 410 mm

DESIGN BASIS

Method: AASHTO 1993 flexible: structural number with layered analysis (AASHTO Guide for Design of Pavement Structures (1993), Part II Ch. 3 (flexible design equation, layered design analysis))
Structural number required 2.91, provided 3.38
- SN over subgrade (Mr = 9000 psi) = 2.91
- Asphalt concrete: SN required above the layer below = 1.87; D = (SN − ΣSN_above)/(a·m) = (1.87 − 0.00)/(0.44·1.00) = 4.24 in → 110 mm (min 75 mm); SN provided so far 1.91
- Aggregate base: SN required above the layer below = 2.43; D = (SN − ΣSN_above)/(a·m) = (2.43 − 1.91)/(0.14·1.00) = 3.73 in → 150 mm (min 150 mm); SN provided so far 2.73
- Aggregate subbase: SN required above the layer below = 2.91; D = (SN − ΣSN_above)/(a·m) = (2.91 − 2.73)/(0.11·1.00) = 1.64 in → 150 mm (min 150 mm); SN provided so far 3.38
Engineer inputs: reliability ZR; overall standard deviation S0; initial/terminal serviceability p0/pt; layer coefficients a_i, drainage m_i, layer moduli
Note: subgrade Mr from Mr(psi) = 1500·CBR (AASHTO 1993 correlation); confirm with tests

SUBGRADE PREPARATION (SCHEMATIC)



Strip topsoil and unsuitable material; proof-roll exposed subgrade.
Scarify and recompact the top 300 mm (dashed) to the specified density.
Replace soft spots identified by proof-rolling (engineer to direct).
Subgrade drains where the geotechnical report requires them.
Density, moisture and CBR testing per the governing specification.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Pavement and subgrade details
Sheet 6 of 19
Scale: 1:10 (layers) (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
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Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

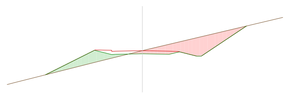
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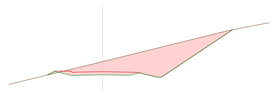
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STA 0+000.00



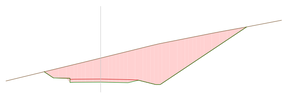
FG 145.58 EG 145.58 cut 18.3 m² fill 7.9 m²
cross slope L -2.00% R -2.00% daylight L 13.6 R 12.7 m

STA 0+009.58



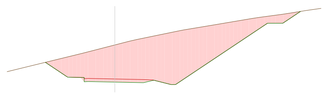
FG 145.35 EG 146.65 cut 45.0 m² fill 0.2 m²
cross slope L -2.00% R -2.00% daylight L 17.0 R 7.3 m

STA 0+020.00



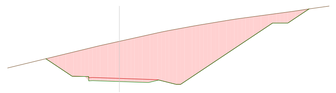
FG 145.10 EG 147.84 cut 77.0 m² fill 0.0 m²
cross slope L -2.00% R -0.60% daylight L 19.1 R 7.4 m

STA 0+040.00



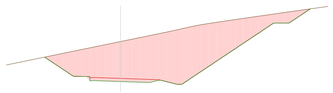
FG 144.62 EG 149.06 cut 124.3 m² fill 0.0 m²
cross slope L -2.08% R +2.08% daylight L 24.3 R 9.1 m

STA 0+049.58



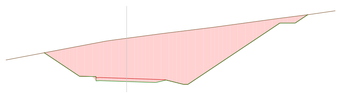
FG 144.39 EG 149.23 cut 137.5 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 24.8 R 9.6 m

STA 0+060.00



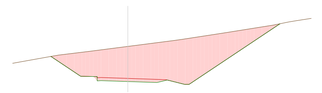
FG 144.14 EG 148.95 cut 137.9 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 24.8 R 9.9 m

STA 0+080.00



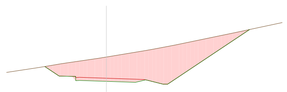
FG 143.66 EG 148.90 cut 137.4 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 23.6 R 10.8 m

STA 0+100.00



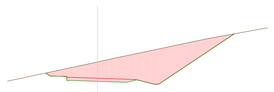
FG 143.19 EG 147.36 cut 107.1 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 19.9 R 10.0 m

STA 0+120.00



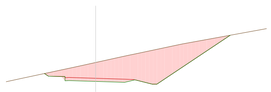
FG 142.71 EG 145.48 cut 74.1 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 18.7 R 8.0 m

STA 0+138.25



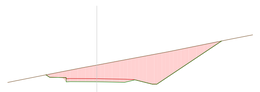
FG 142.27 EG 144.40 cut 60.7 m² fill 0.0 m²
cross slope L -3.36% R +3.36% daylight L 17.9 R 6.8 m

STA 0+140.00



FG 142.23 EG 144.29 cut 58.9 m² fill 0.0 m²
cross slope L -3.13% R +3.13% daylight L 17.6 R 6.7 m

STA 0+160.00



FG 140.88 EG 142.65 cut 48.5 m² fill 0.0 m²
cross slope L -2.00% R +0.45% daylight L 16.3 R 6.6 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 7 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

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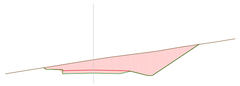
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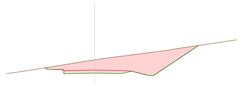
CROSS SECTIONS 1:400 (red = cut, green = fill, green line = subgrade datum)

STA 0+178.25



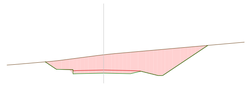
FG 139.57 EG 140.92 cut 34.4 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 13.8 R 6.5 m

STA 0+180.00



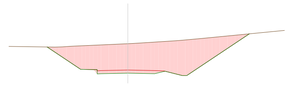
FG 139.45 EG 140.77 cut 33.4 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 13.6 R 6.5 m

STA 0+200.00



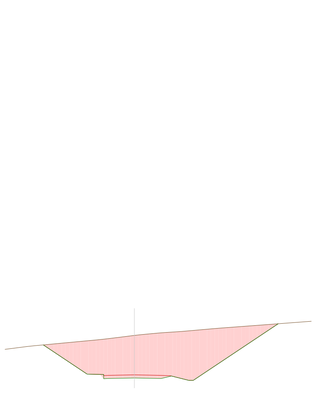
FG 138.01 EG 140.00 cut 43.9 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 13.6 R 7.6 m

STA 0+220.00



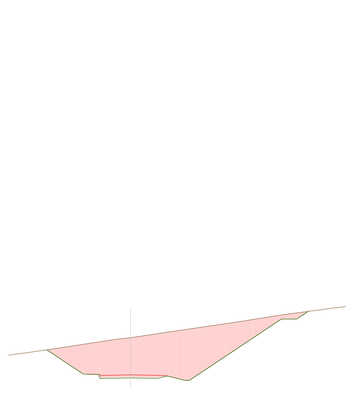
FG 136.57 EG 140.00 cut 79.3 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 15.9 R 10.5 m

STA 0+240.00



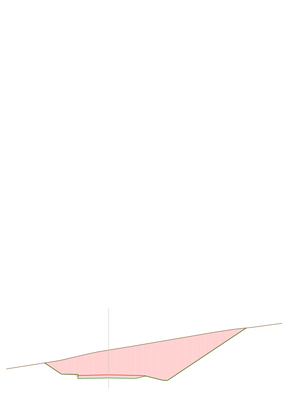
FG 135.13 EG 140.28 cut 125.5 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 18.8 R 11.9 m

STA 0+260.00



FG 133.69 EG 138.54 cut 128.7 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 23.2 R 10.9 m

STA 0+276.41



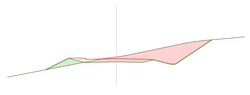
FG 132.50 EG 135.74 cut 80.5 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 18.0 R 8.4 m

STA 0+280.00



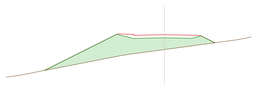
FG 132.26 EG 134.96 cut 66.9 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 16.7 R 7.7 m

STA 0+291.41



FG 131.65 EG 131.87 cut 18.5 m² fill 2.3 m²
cross slope L -2.00% R -2.00% daylight L 12.5 R 9.2 m

STA 0+300.00



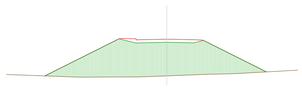
FG 131.28 EG 129.34 cut 0.0 m² fill 32.9 m²
cross slope L -2.00% R -2.00% daylight L 6.6 R 15.7 m

STA 0+320.00



FG 130.43 EG 124.44 cut 0.0 m² fill 142.7 m²
cross slope L -2.00% R -2.00% daylight L 12.2 R 25.7 m

STA 0+340.00



FG 129.58 EG 124.78 cut 0.0 m² fill 91.9 m²
cross slope L -2.00% R -2.00% daylight L 13.0 R 15.9 m

CROSS SECTIONS 1:400 (red = cut, green = fill, green line = subgrade datum)

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 8 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

NOT FOR CONSTRUCTION

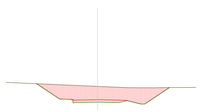
Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

ENGINEER SEAL / SIGNATURE

STA 0+360.00

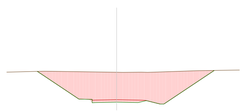
FG 128.72 EG 127.60 cut 0.0 m² fill 10.2 m²
cross slope L -2.00% R -2.00% daylight L 6.9 R 8.2 m

STA 0+380.00



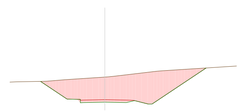
FG 127.87 EG 130.04 cut 44.6 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 11.7 R 9.9 m

STA 0+400.00



FG 127.02 EG 131.49 cut 104.5 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 15.9 R 12.9 m

STA 0+420.00



FG 126.17 EG 129.85 cut 88.4 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 16.5 R 10.5 m

STA 0+440.00

FG 125.17 EG 126.64 cut 36.0 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 13.6 R 6.8 m

STA 0+460.00

FG 124.11 EG 122.28 cut 0.0 m² fill 30.3 m²
cross slope L -2.00% R -2.00% daylight L 6.0 R 14.6 m

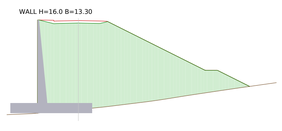
STA 0+480.00

FG 123.05 EG 116.32 cut 0.0 m² fill 184.3 m²
cross slope L -2.00% R -2.00% daylight L 14.0 R 29.4 m

STA 0+500.00

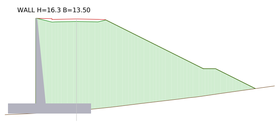
FG 121.99 EG 109.83 cut 0.0 m² fill 234.0 m²
cross slope L -2.00% R -2.00% daylight L 23.5 R 6.7 m

STA 0+515.33



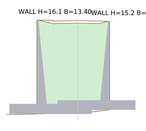
FG 121.18 EG 106.67 cut 0.0 m² fill 319.4 m²
cross slope L -2.00% R -2.00% daylight L 28.0 R 6.7 m

STA 0+520.00



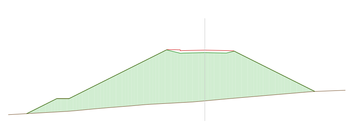
FG 120.94 EG 105.97 cut 0.0 m² fill 339.7 m²
cross slope L -2.00% R -2.00% daylight L 29.2 R 6.7 m

STA 0+540.00



FG 120.12 EG 105.05 cut 0.0 m² fill 174.1 m²
cross slope L -2.00% R -2.00% daylight L 5.2 R 6.7 m

STA 0+560.00



FG 119.63 EG 111.35 cut 0.0 m² fill 232.8 m²
cross slope L -2.00% R -2.00% daylight L 18.0 R 29.1 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 9 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

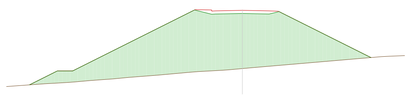
NOT FOR CONSTRUCTION

Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

ENGINEER SEAL / SIGNATURE

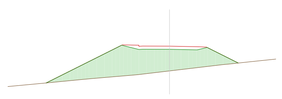
CROSS SECTIONS 1:500 (red = cut, green = fill, green line = subgrade datum)

STA 0+562.02



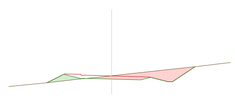
FG 119.60 EG 111.99 cut 0.0 m² fill 204.2 m²
cross slope L -2.00% R -2.00% daylight L 16.8 R 27.8 m

STA 0+580.00



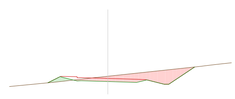
FG 119.47 EG 116.23 cut 0.0 m² fill 57.7 m²
cross slope L -2.00% R +0.26% daylight L 9.0 R 16.1 m

STA 0+600.00



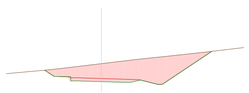
FG 119.63 EG 119.66 cut 10.4 m² fill 2.0 m²
cross slope L -2.77% R +2.77% daylight L 10.9 R 8.5 m

STA 0+602.02



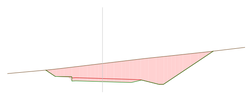
FG 119.67 EG 119.95 cut 14.3 m² fill 1.1 m²
cross slope L -3.03% R +3.03% daylight L 11.3 R 7.8 m

STA 0+620.00



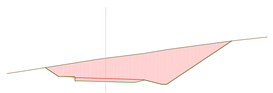
FG 120.13 EG 122.07 cut 45.7 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 14.4 R 7.4 m

STA 0+620.33



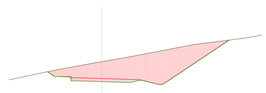
FG 120.14 EG 122.09 cut 46.1 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 14.5 R 7.4 m

STA 0+640.00



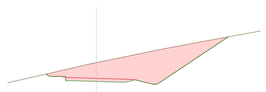
FG 120.79 EG 123.44 cut 64.0 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 16.5 R 7.9 m

STA 0+660.00



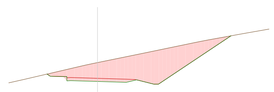
FG 121.45 EG 123.66 cut 58.1 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 16.6 R 7.1 m

STA 0+674.78



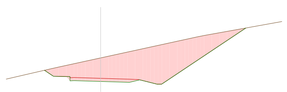
FG 121.95 EG 124.00 cut 56.6 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 17.2 R 6.6 m

STA 0+680.00



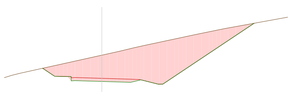
FG 122.10 EG 124.15 cut 57.4 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 17.4 R 6.6 m

STA 0+700.00



FG 122.27 EG 124.93 cut 74.7 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 19.0 R 7.3 m

STA 0+711.01



FG 122.09 EG 125.11 cut 85.1 m² fill 0.0 m²
cross slope L -3.03% R +3.03% daylight L 19.9 R 7.7 m

CROSS SECTIONS 1:400 (red = cut, green = fill, green line = subgrade datum)

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 10 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

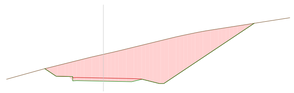
NOT FOR CONSTRUCTION

NOT FOR CONSTRUCTION

Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

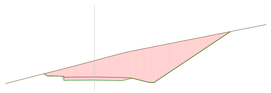
ENGINEER SEAL / SIGNATURE

STA 0+720.00



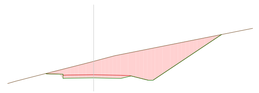
FG 121.79 EG 124.88 cut 86.7 m² fill 0.0 m²
cross slope L -2.00% R +1.90% daylight L 19.7 R 7.6 m

STA 0+740.00



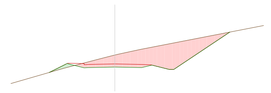
FG 120.67 EG 122.81 cut 60.7 m² fill 0.0 m²
cross slope L -2.00% R -0.62% daylight L 17.7 R 6.6 m

STA 0+744.78



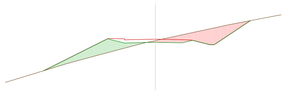
FG 120.31 EG 122.06 cut 50.4 m² fill 0.0 m²
cross slope L -2.00% R -1.22% daylight L 16.7 R 6.2 m

STA 0+751.01



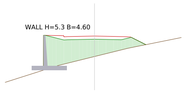
FG 119.81 EG 120.97 cut 36.1 m² fill 0.8 m²
cross slope L -2.00% R -2.00% daylight L 15.0 R 8.6 m

STA 0+760.00



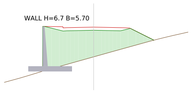
FG 119.09 EG 118.90 cut 13.7 m² fill 10.9 m²
cross slope L -2.00% R -2.00% daylight L 12.4 R 14.6 m

STA 0+780.00



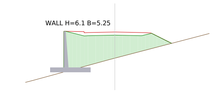
FG 117.49 EG 114.82 cut 0.0 m² fill 30.4 m²
cross slope L -2.00% R -2.00% daylight L 6.7 R 6.7 m

STA 0+800.00



FG 115.89 EG 112.01 cut 0.0 m² fill 46.9 m²
cross slope L -2.00% R -2.00% daylight L 7.9 R 6.7 m

STA 0+820.00



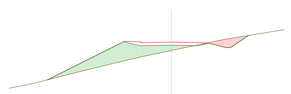
FG 114.29 EG 110.88 cut 0.0 m² fill 40.5 m²
cross slope L -2.00% R -2.00% daylight L 7.4 R 6.7 m

STA 0+840.00



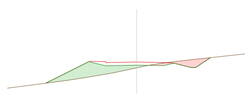
FG 112.69 EG 110.59 cut 0.0 m² fill 53.3 m²
cross slope L -2.00% R -2.00% daylight L 5.8 R 26.4 m

STA 0+860.00



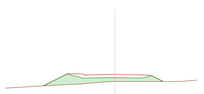
FG 111.21 EG 110.11 cut 3.6 m² fill 22.1 m²
cross slope L -2.00% R -2.00% daylight L 10.1 R 16.2 m

STA 0+880.00



FG 109.89 EG 109.12 cut 3.8 m² fill 12.4 m²
cross slope L -2.00% R -2.00% daylight L 9.6 R 11.9 m

STA 0+900.00



FG 108.57 EG 107.64 cut 0.0 m² fill 9.9 m²
cross slope L -2.00% R -2.00% daylight L 6.3 R 9.3 m

CROSS SECTIONS 1:400 (red = cut, green = fill, green line = subgrade datum)

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 11 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

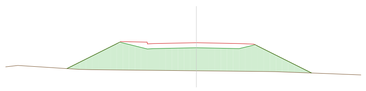
NOT FOR CONSTRUCTION

NOT FOR CONSTRUCTION

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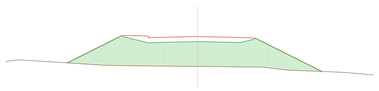
ENGINEER SEAL / SIGNATURE

STA 0+920.00



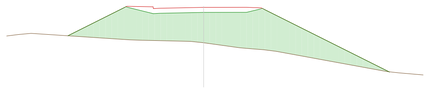
FG 107.25 EG 104.96 cut 0.0 m² fill 30.9 m²
cross slope L -2.00% R -2.00% daylight L 9.4 R 10.6 m

STA 0+922.83



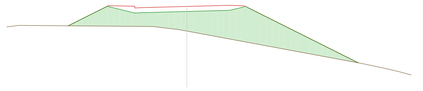
FG 107.06 EG 104.63 cut 0.0 m² fill 34.3 m²
cross slope L -2.00% R -2.00% daylight L 10.2 R 10.6 m

STA 0+940.00



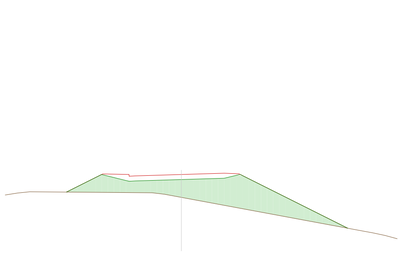
FG 105.93 EG 103.04 cut 0.0 m² fill 52.0 m²
cross slope L +0.16% R -2.00% daylight L 15.2 R 11.1 m

STA 0+960.00



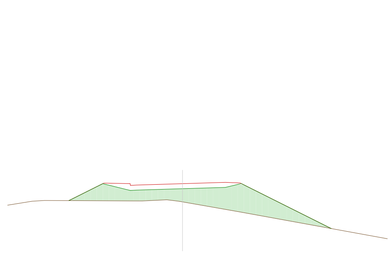
FG 104.61 EG 102.61 cut 0.0 m² fill 34.7 m²
cross slope L +2.67% R -2.67% daylight L 14.0 R 9.7 m

STA 0+962.83



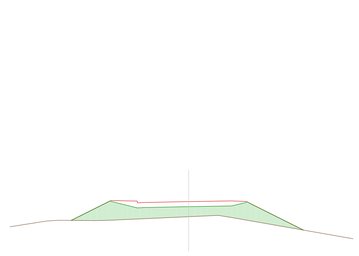
FG 104.43 EG 102.56 cut 0.0 m² fill 31.3 m²
cross slope L +3.03% R -3.03% daylight L 13.6 R 9.4 m

STA 0+968.80



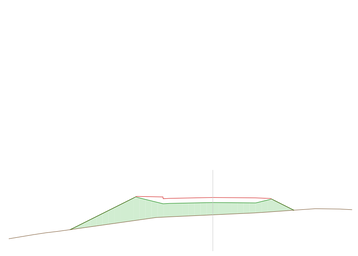
FG 104.03 EG 102.55 cut 0.0 m² fill 24.6 m²
cross slope L +3.03% R -3.03% daylight L 12.1 R 9.3 m

STA 0+980.00



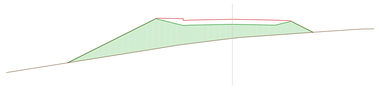
FG 103.30 EG 102.06 cut 0.0 m² fill 16.7 m²
cross slope L +1.62% R -2.00% daylight L 9.4 R 9.6 m

STA 1+000.00



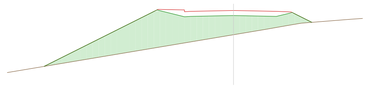
FG 102.04 EG 100.63 cut 0.0 m² fill 18.1 m²
cross slope L -0.89% R -2.00% daylight L 6.6 R 11.6 m

STA 1+008.80



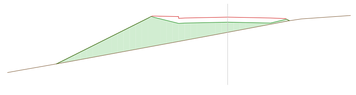
FG 101.56 EG 100.00 cut 0.0 m² fill 24.2 m²
cross slope L -2.00% R -2.00% daylight L 6.6 R 13.4 m

STA 1+020.00



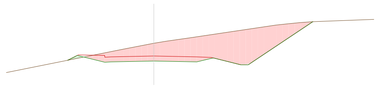
FG 100.94 EG 98.96 cut 0.0 m² fill 33.8 m²
cross slope L -2.00% R -2.00% daylight L 6.4 R 15.5 m

STA 1+040.00



FG 99.83 EG 98.59 cut 0.0 m² fill 20.4 m²
cross slope L -2.00% R -2.00% daylight L 5.0 R 14.0 m

STA 1+060.00



FG 98.72 EG 99.76 cut 28.9 m² fill 0.2 m²
cross slope L -2.00% R -2.00% daylight L 13.0 R 7.0 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 12 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

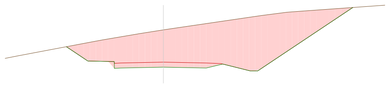
NOT FOR CONSTRUCTION

Not reviewed or sealed by a licensed engineer
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ENGINEER SEAL / SIGNATURE

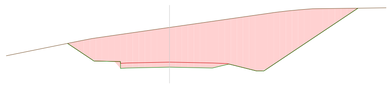
CROSS SECTIONS 1:250 (red = cut, green = fill, green line = subgrade datum)

STA 1+080.00



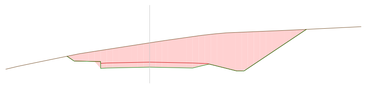
FG 97.61 EG 100.25 cut 61.9 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 15.5 R 7.9 m

STA 1+100.00



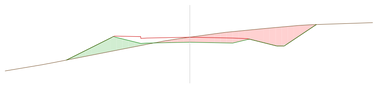
FG 96.51 EG 99.34 cut 66.5 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 15.4 R 8.3 m

STA 1+120.00



FG 95.40 EG 96.98 cut 36.2 m² fill 0.0 m²
cross slope L -2.00% R -2.00% daylight L 12.8 R 6.7 m

STA 1+135.65



FG 94.53 EG 94.53 cut 9.8 m² fill 3.8 m²
cross slope L -2.00% R -2.00% daylight L 10.3 R 10.1 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Cross sections
Sheet 13 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

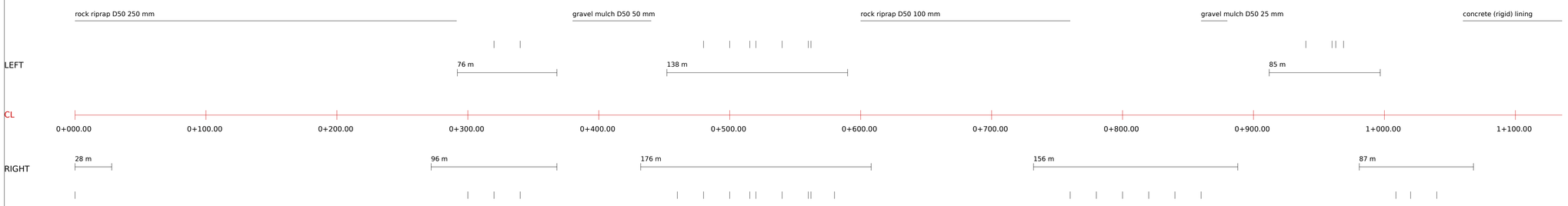
NOT FOR CONSTRUCTION

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Not reviewed or sealed by a licensed engineer.

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CROSS SECTIONS 1:250 (red = cut, green = fill, green line = subgrade datum)

Ticks: hazards inside the clear zone (embankment above warrant height, steep slopes, walls). Thick lines: barrier runs incl. length of need. Blue: ditch lining reaches.



Side	From	To	Length m	LON m	Type	Reason
left	0+292.00	0+368.00	76	28.0	W-beam guardrail	critical fill slope
left	0+452.00	0+590.02	138	28.0	W-beam guardrail	critical fill slope; retaining wall drop-off
left	0+912.00	0+996.80	85	28.0	W-beam guardrail	critical fill slope
right	0+000.00	0+028.00	28	28.0	W-beam guardrail	critical fill slope
right	0+272.00	0+368.00	96	28.0	W-beam guardrail	critical fill slope
right	0+432.00	0+608.00	176	28.0	W-beam guardrail	critical fill slope; retaining wall drop-off
right	0+732.00	0+888.00	156	28.0	W-beam guardrail	critical fill slope; retaining wall drop-off
right	0+980.80	1+068.00	87	28.0	W-beam guardrail	critical fill slope

Side	From	To	Q m3/s	Depth m	V m/s	Slope %	tau_d N/m2	Lining	tau_p N/m2
left	0+000.00	0+291.41	0.000	0.00	1.49	7277.59	221.3	rock riprap D50 250 mm	297
left	0+380.00	0+440.00	0.017	0.05	0.45	5.30	26.1	gravel mulch D50 50 mm	38
left	0+600.00	0+760.00	0.044	0.08	0.67	8.00	63.5	rock riprap D50 100 mm	76
left	0+860.00	0+880.00	0.006	0.02	0.36	6.59	15.0	gravel mulch D50 25 mm	19
left	1+060.00	1+135.65	0.021	0.00	11.13	4729.39	1439.2	concrete (rigid) lining	rigid

Clear zone 5 m (engineer input from AASHTO Roadside Design Guide Table 3-1 for the design speed, ADT and slopes (demo value)); recoverable ≤ 1V:4H, traversable ≤ 1V:3H, embankment warrant height 3 m: engineer inputs.
right 10: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe
left 300: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe
left 360: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe
right 360: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe
left 460: non-recoverable fill slope 1V:2.0H inside the clear zone; clear run-out area needed at the toe

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Roadside protection layout (straight-line diagram)
Sheet 14 of 19
Scale: H 1:1800 (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

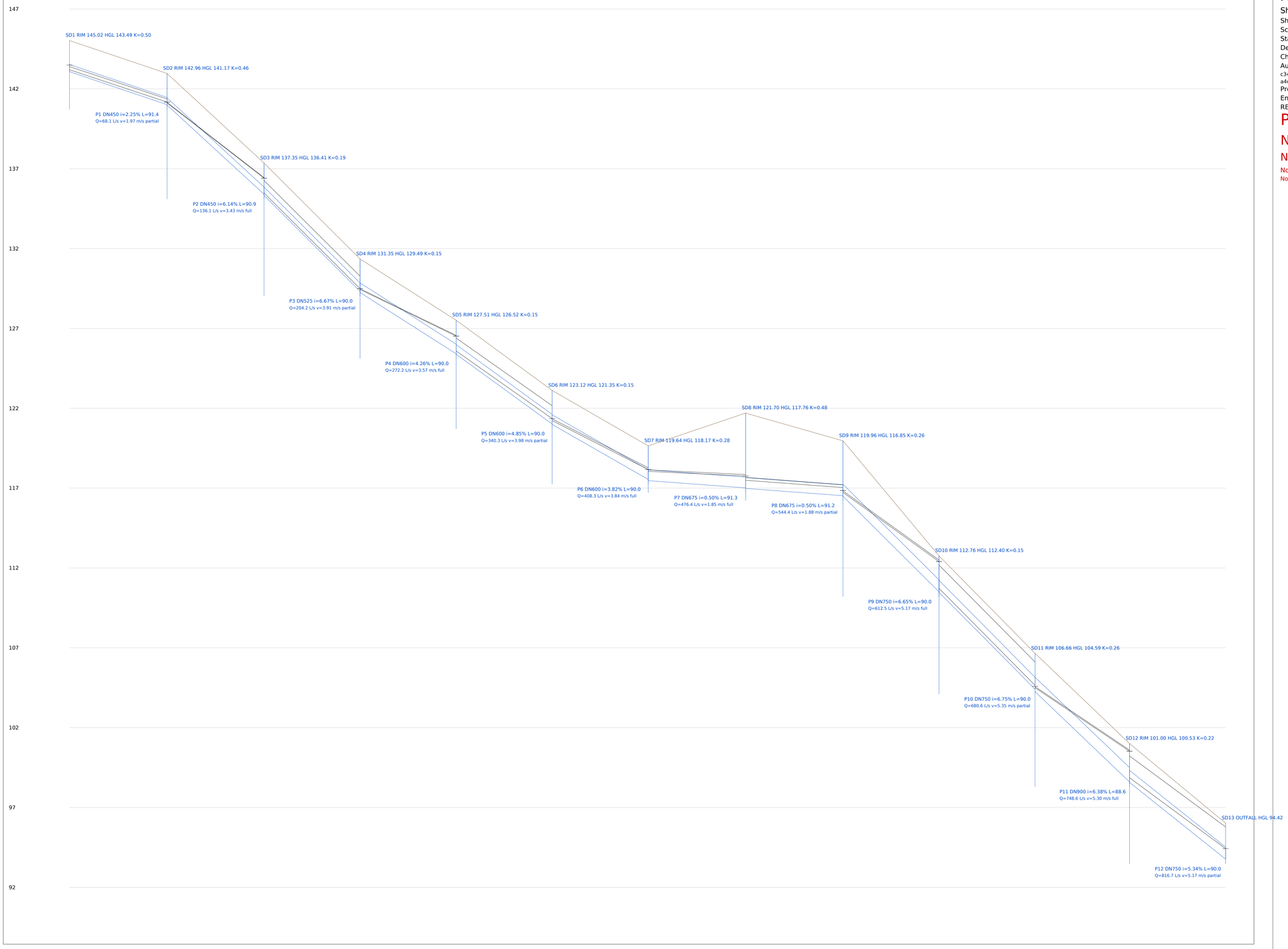
NOT FOR CONSTRUCTION

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PIPE PROFILE - Storm H 1:1773 V 1:119

cyan = HGL, red = EGL, brown = rim; method FHWA HEC-22 approximate method: outfall HGL = max(tailwater, invert + (dc+D)/2); full flow: HGL_up = HGL_dn + Sf·L, Sf = (Qn/(A R^(2/3)))²; partial flow: normal depth; structure: E = EGL_out + K·Vo²/2g, K by deflection (0°0.15, 22.5°0.45, 45°0.75, 60°0.85, 90°1.0; start structure 0.5). Coefficients transcribed — verify against the HEC-22 edition in use.



P-SURCH advisory pipe P2 Pipe flows full (surcharged): 0.554 vs max 0.000 m
P-SURCH advisory pipe P4 Pipe flows full (surcharged): 0.505 vs max 0.000 m
P-SURCH advisory pipe P6 Pipe flows full (surcharged): 0.023 vs max 0.000 m
P-SURCH advisory pipe P7 Pipe flows full (surcharged): 0.070 vs max 0.000 m
P-SURCH advisory pipe P9 Pipe flows full (surcharged): 1.146 vs max 0.000 m
P-SURCH advisory pipe P11 Pipe flows full (surcharged): 1.022 vs max 0.000 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Pipe profile Storm
Sheet 15 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

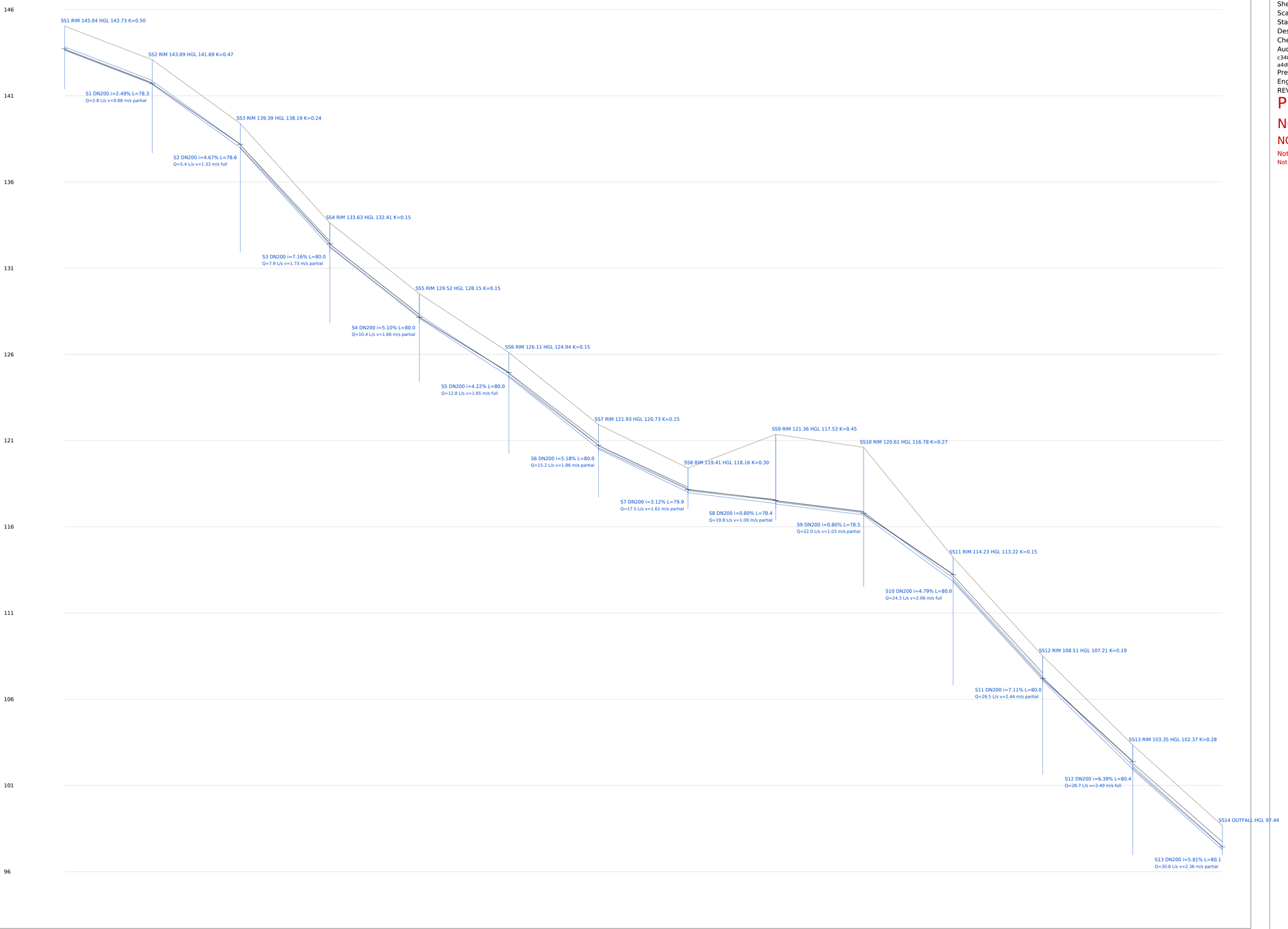
NOT FOR CONSTRUCTION

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Not reviewed or sealed by a licensed engineer.

ENGINEER SEAL / SIGNATURE

PIPE PROFILE - Sanitary H 1:1693 V 1:110

cyan = HGL, red = EGL, brown = rim; method FHWA HEC-22 approximate method: outfall HGL = max(tailwater, invert + (dc+D)/2); full flow: HGL_up = HGL_dn + Sf·L, Sf = (Qn/(A R^(2/3)))²; partial flow: normal depth; structure: E = EGL_out + K·Vo²/2g, K by deflection (0°0.15, 22.5°0.45, 45°0.75, 60°0.85, 90°1.0; start structure 0.5). Coefficients transcribed — verify against the HEC-22 edition in use.



P-SURCH advisory pipe S2 Pipe flows full (surcharged): 0.001 vs max 0.000 m
P-SURCH advisory pipe S5 Pipe flows full (surcharged): 0.035 vs max 0.000 m
P-SURCH advisory pipe S10 Pipe flows full (surcharged): 0.195 vs max 0.000 m
P-SURCH advisory pipe S12 Pipe flows full (surcharged): 0.221 vs max 0.000 m

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Pipe profile Sanitary
Sheet 16 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

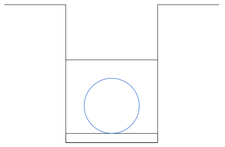
NOT FOR CONSTRUCTION

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Not reviewed or sealed by a licensed engineer.

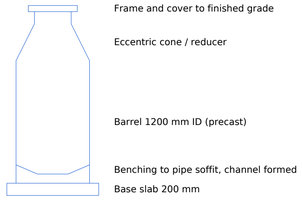
ENGINEER SEAL / SIGNATURE

PIPE TRENCH (SCHEMATIC, 1:50)

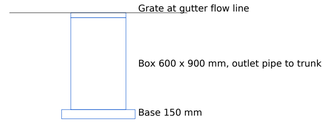


Trench width D + 600 mm = 1.50 m

MANHOLE (SCHEMATIC, 1:50)



GRATE INLET (SCHEMATIC, 1:50)



Network	Structure	Rim	Sump invert	Depth m	Inverts out (pipe, D mm)
Storm	SD1	145.024	143.074	1.95	P1 143.074 D450
Storm	SD2	142.963	140.983	1.98	P2 140.983 D450
Storm	SD3	137.352	135.327	2.03	P3 135.327 D525
Storm	SD4	131.346	129.246	2.10	P4 129.246 D600
Storm	SD5	127.512	125.382	2.13	P5 125.382 D600
Storm	SD6	123.116	120.986	2.13	P6 120.986 D600
Storm	SD7	119.643	117.468	2.17	P7 117.468 D675
Storm	SD8	121.698	116.981	4.72	P8 116.981 D675
Storm	SD9	119.961	116.495	3.47	P9 116.495 D750
Storm	SD10	112.756	110.476	2.28	P10 110.476 D750
Storm	SD11	106.655	104.255	2.40	P11 104.255 D900
Storm	SD12	101.004	98.574	2.43	P12 98.574 D750
Storm	SD13	96.020	93.770	2.25	
Sanitary	SS1	145.038	143.638	1.40	S1 143.638 D200
Sanitary	SS2	143.085	141.655	1.43	S2 141.655 D200
Sanitary	SS3	139.385	137.955	1.43	S3 137.955 D200
Sanitary	SS4	133.626	132.196	1.43	S4 132.196 D200
Sanitary	SS5	129.515	128.085	1.43	S5 128.085 D200
Sanitary	SS6	126.107	124.677	1.43	S6 124.677 D200
Sanitary	SS7	121.929	120.499	1.43	S7 120.499 D200
Sanitary	SS8	119.407	117.977	1.43	S8 117.977 D200
Sanitary	SS9	121.364	117.320	4.04	S9 117.320 D200
Sanitary	SS10	120.611	116.662	3.95	S10 116.662 D200
Sanitary	SS11	114.230	112.800	1.43	S11 112.800 D200
Sanitary	SS12	108.509	107.079	1.43	S12 107.079 D200
Sanitary	SS13	103.345	101.915	1.43	S13 101.915 D200
Sanitary	SS14	98.662	97.262	1.40	

Details are schematic. Final dimensions, materials, frames, grates, joint types and bedding class per the agency standard drawings and the governing specification cited on the standard-details sheet. Structure schedule lists every structure designed by the pipe-network tool (first 39 rows).

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Drainage details and structure schedule

Sheet 17 of 19

Scale: 1:50 (details) (A1)

Standards: Us

Design speed: 50 km/h

Checks: 218/218 pass

Audit log head:

c340436ef0da561e02ee22ad660b07ad

a4df5d59149906780694e48e0e9c0ea5

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

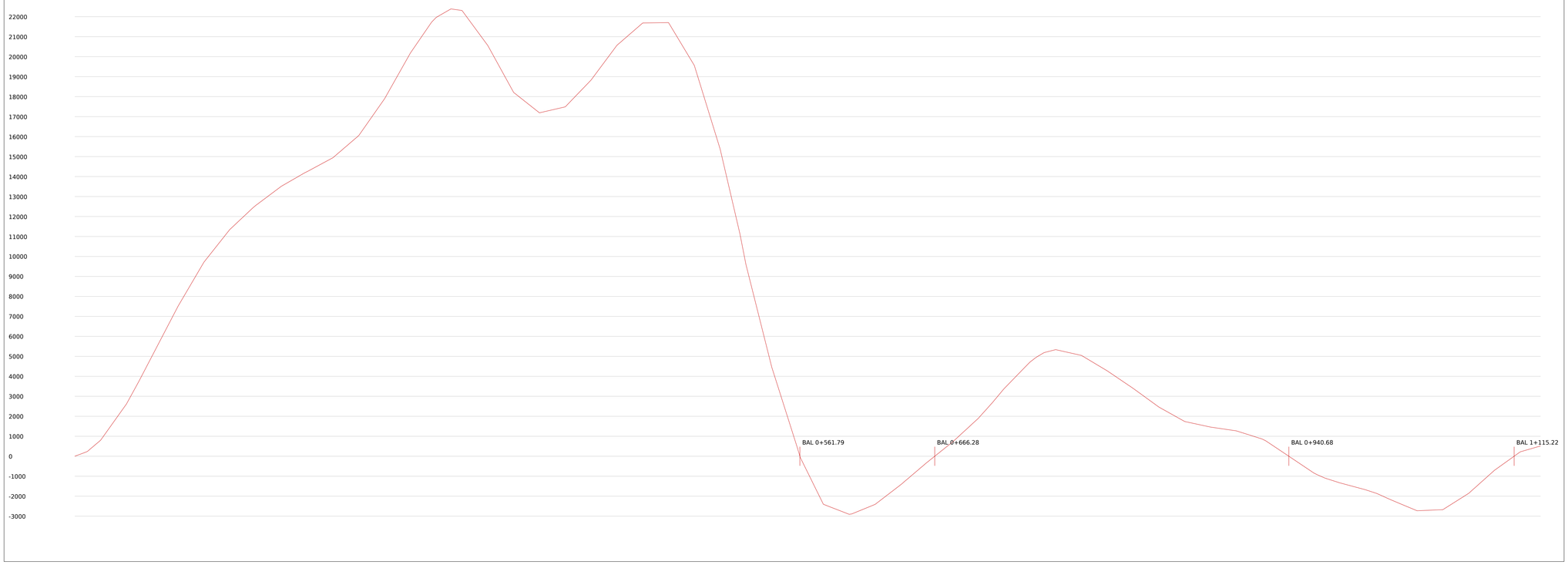
NOT FOR CONSTRUCTION

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MASS HAUL (m³) cut factor 0.90 final ordinate 514 m³



Station range	Cut m³	Fill m³	Asphalt concrete m³	Aggregate base m³	Aggregate subbase m³
0+000.00–0+100.00	10838	40	84.9	115.7	115.7
0+100.00–0+200.00	5811	0	84.4	115.2	115.2
0+200.00–0+300.00	8370	164	82.5	112.5	112.5
0+300.00–0+400.00	1936	5224	82.5	112.5	112.5
0+400.00–0+500.00	3533	6632	82.5	112.5	112.5
0+500.00–0+600.00	104	18378	83.0	113.2	113.2
0+600.00–0+700.00	5350	13	85.4	116.4	116.4
0+700.00–0+800.00	4022	1242	83.4	113.7	113.7
0+800.00–0+900.00	148	3135	82.5	112.5	112.5
0+900.00–1+000.00	0	2947	83.8	114.3	114.3
1+000.00–1+100.00	2480	1258	82.5	112.5	112.5
1+100.00–1+200.00	1386	30	29.4	40.1	40.1
Total	43976	39064	946.8	1291.1	1291.1

Pipe schedule

RCP DN450: 182.3 m
RCP DN525: 90.0 m
RCP DN600: 270.0 m
RCP DN675: 182.5 m
RCP DN750: 270.0 m
RCP DN900: 88.6 m
Structures: 13

Pipe schedule

PVC DN200: 1034.3 m
Structures: 14

Grading yard: cut 275 m³ fill 1006 m³ (grid 0.25 m)

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Mass haul and quantities
Sheet 18 of 19
Scale: AS SHOWN (A1)
Standards: Us
Design speed: 50 km/h
Checks: 218/218 pass
Audit log head:
c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5
Prepared by: civil-mcp agent tools
Engineer in responsible charge:
REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

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Not reviewed or sealed by a licensed engineer
Not reviewed or sealed by a licensed engineer.

ENGINEER SEAL / SIGNATURE

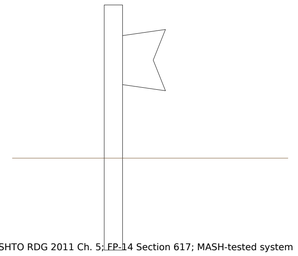
D1 Curb and gutter



Reference: State DOT standard plan; FP-14 Section 609

Schematic only - not to scale. Use the referenced standard drawing.

D2 W-beam guardrail



Reference: AASHTO RDG 2011 Ch. 5; FP-14 Section 617; MASH-tested system

Schematic only - not to scale. Use the referenced standard drawing.

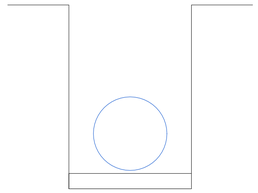
D3 Riprap / ditch lining



Reference: HEC-15 (2005); FP-14 Sections 251, 629

Schematic only - not to scale. Use the referenced standard drawing.

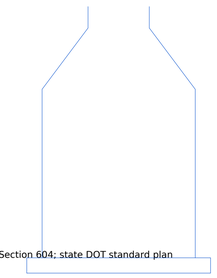
D4 Pipe trench and bedding



Reference: FP-14 Section 602; pipe manufacturer / AASHTO LRFD Sec. 12

Schematic only - not to scale. Use the referenced standard drawing.

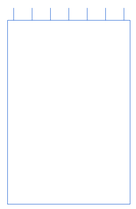
D5 Manhole



Reference: FP-14 Section 604; state DOT standard plan

Schematic only - not to scale. Use the referenced standard drawing.

D6 Grate inlet



Reference: HEC-22 (2009) inlet capacity; FP-14 Section 604

Schematic only - not to scale. Use the referenced standard drawing.

D7 Pavement edge / shoulder



Reference: AASHTO 93 design; FP-14 Sections 301, 401

Schematic only - not to scale. Use the referenced standard drawing.

D8 Sidewalk



Reference: FP-14 Section 615; PROWAG accessibility

Schematic only - not to scale. Use the referenced standard drawing.

Demo rural collector (USGS 3DEP terrain, Palo Alto CA)

Sheet: Standard details atlas (references)

Sheet 19 of 19

Scale: NTS (A1)

Standards: Us

Design speed: 50 km/h

Checks: 218/218 pass

Audit log head:

c340436ef0da561e02ee22ad660b07ad
a4df5d59149906780694e48e0e9c0ea5

Prepared by: civil-mcp agent tools

Engineer in responsible charge:

REVIEWER-PLACEHOLDER

PRELIMINARY

NOT FOR CONSTRUCTION

NOT FOR CONSTRUCTION

Not reviewed or sealed by a licensed engineer

Not reviewed or sealed by a licensed engineer.

ENGINEER SEAL / SIGNATURE